Grounding the Loop on Both Sides: Interaction as a Third Test-Time Compute Axis, and Why Its Gains Are Invisible Without Grounded Evaluation

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Abstract

Test-time compute is dominated by two scaling axes that both operate inside a model's own token space: *reasoning* (longer chains of thought) and *sampling* (best-of-N). Both are bounded by the information already in the weights and prompt, a limit that follows from the data-processing inequality (DPI). A third axis, *interaction* with an environment that returns grounded feedback, can exceed that bound by importing information from outside the model. We argue that one variable, **grounding**, governs this axis, and that it must hold on *both* sides of the loop: grounded *feedback* drives improvement past the reasoning/sampling ceiling, and grounded *evaluation* is required even to *measure* it. We formalize the first claim as a DPI argument over a Type 0–3 feedback taxonomy, and establish the second on rendered visual artifacts: the default vision–language-model (VLM) judge is structurally blind to the geometric defects the loop fixes, rating 14 of 15 single-shot academic figures "perfect" when only 3 are actually clean, and a VLM *reviewer* even makes layouts worse. A deterministic DOM-geometry instrument (overlap, clipping, overflow, and misalignment, measured exactly from the rendered layout) instead reveals 40–74% defect reductions under grounded feedback across **four** visual modalities measured identically (academic figures, dense slides, responsive web pages, and SVG/CSS animations), with every paired sign test at $p < 0.002$ and every bootstrap 95% CI excluding zero.

On execution-grounded code the harness lifts a frontier model from a two-thirds pass-rate to a strict 100%, an in-modality control isolates the *execution* of tests, not the extra reviewing pass, as the cause, and the effect replicates across three model families (three seeds each) and a held-out suite; the geometry result reproduces with Gemini 3.1 Pro as proposer under the same model-free instrument. At a fixed token budget, reasoning-only and best-of-N saturate while every interaction strategy approaches a perfect pass-rate, the proposer–reviewer harness doing so at the lowest token cost with zero seed variance. Finally, the harness's behavior partially distills into an 8B student. Interaction scaling is real, distinct, and predictable, but its gains are visible only when both the feedback and the metric are grounded.

Code, task suites, prompts, and deterministic instruments:
https://github.com/19PINE-AI/interaction-scaling

Website (with an interactive single-shot-vs-reviewed results browser):
https://01.me/research/interaction-scaling

1 Introduction

Scaling test-time compute has become the dominant lever for improving large models on artifact-producing tasks: code, web pages, slides, figures, animations, video edits, research reports. The literature has formalized two axes, and both operate *inside the model's own token space*. *Reasoning scaling* [4, 19, 25, 26] spends more tokens thinking before committing to an answer. *Sampling scaling* [1, 24] draws more independent attempts and selects one. In both, every additional token is a function of the same frozen weights and the same fixed prompt: more compute reorganizes information already in the system but imports none.

A third axis breaks this closure: *interaction* with an external environment that returns **grounded** feedback, such as a test result, a rendered layout, or a search verdict. Such feedback is not a function of the weights. It is an observation of the artifact's real behavior, and it can carry information the model never had. This axis is well established empirically [6, 17, 18, 27], but accounts of *why* and *when* it should beat reasoning- or sampling-only scaling remain informal and, we argue, half-stated. Practitioner reports at industrial scale sharpen the stakes: in a controlled 900-run deployment study at ByteDance, frontier coding models pass the functional-correctness bar yet miss on delivery quality until a feedback "harness" is added, and that report had to invent a multi-axis quality metric before the gap was even visible [7]: a field observation of exactly the two-sided grounding problem we formalize below (detailed in Section 8).

Grounding is the variable, on both sides of the loop. Our thesis is that one variable governs interaction scaling: **grounding**, the property that a signal is computed from the artifact's actual behavior rather than from a model's opinion of it. And grounding must hold on *both* sides of the proposer–reviewer loop (Figure 1):

1. **Grounded feedback** (the reviewer signal that drives revision): run the code and read `pytest`, render the HTML and measure its bounding boxes, or issue the query and read the result. This is the side prior work emphasizes, and a data-processing-inequality argument (Section 2) shows it is exactly what lets interaction exceed the ceiling that bounds reasoning and sampling.
2. **Grounded evaluation** (the metric that *measures* the gain): the same observation, used as the score. This side is almost always overlooked, and getting it wrong silently hides the effect.

The second point is not a detail. The default evaluator for visual artifacts is a vision–language model (VLM) reading a screenshot, and it is *structurally* blind to the defects that matter: content overflowing the canvas is cropped off-frame before the image reaches the judge, and small overlaps fall below its acuity. On dense academic figures this VLM judge rates 14 of 15 single-shot renders "perfect" (an initial 15-figure probe) when a deterministic check finds only 3 are actually clean (Section 5). Worse, when the same VLM is used as the *reviewer*, its edits make slide geometry *worse*. Ungrounded on either side, the third axis either does not fire or cannot be seen.

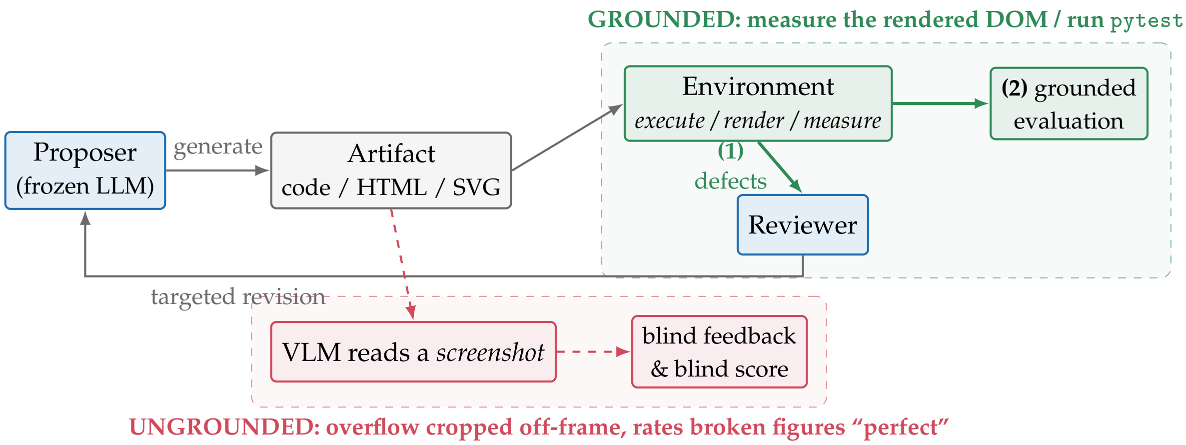


Figure 1. Grounding must hold on both sides of the interaction loop. A proposer emits an artifact, and a grounded environment *executes/renders/measures* it. That same observation does double duty. **(1)** As grounded *feedback*, it becomes a defect list the reviewer turns into a targeted revision, the channel that lets interaction import information from outside the weights and exceed the reasoning/sampling ceiling (Section 2). **(2)** As grounded *evaluation*, it is the score that makes the gain measurable. Substituting the default VLM-on-a-screenshot judge (red lane) breaks *both*: it is structurally blind to off-canvas overflow and fine overlap, so it neither scores the defect nor (as a reviewer) fixes it.

Contributions.

1. **A grounding framework (Section 2).** A Type 0–3 feedback taxonomy and a data-processing-inequality argument that motivates why ungrounded review and reasoning (and best-of-N selection) saturate while grounded feedback need not. We add the symmetric observation that ungrounded *evaluation* cannot measure the gain. Beyond the documented *biases* of model-as-judge evaluation [28], we isolate a sharper failure: for artifacts whose quality is a measurable physical property, a screenshot judge is *structurally blind* (not merely noisy), which silently hides a real interaction-scaling effect.

2. **Interaction is a distinct axis that clears the in-token-space ceiling (Section 4).** At a matched token budget, reasoning and best-of-N saturate while every interaction strategy approaches a perfect pass-rate. An in-modality control isolates test *execution* as the active ingredient, and the harness is the most token-efficient, zero-variance strategy.

3. **Grounded evaluation, and a deterministic instrument (Section 5).** We show the VLM judge is structurally blind, build a deterministic DOM-geometry+alignment instrument, and report large, significant, CI-backed defect reductions under grounded feedback across four visual modalities measured identically (gains the standard metric cannot even see), plus a controlled ablation in which VLM feedback makes layout *worse*.

4. **Robustness and internalization (Sections 6 and 7).** The effects replicate across model families (including grounded geometry on Gemini 3.1 Pro), to held-out tasks, and under a propose-heavy budget allocation. The harness's behavior also partially distills into an 8B student at $\sim$10$\times$ lower deployment cost.

2 The Grounded-Feedback Framework

Our contribution is not a new self-correction method but an account of *why* interaction is a separate axis, *when* it helps (a feedback-actionability taxonomy), and a previously-missing requirement: that the *evaluation* be grounded too, without which the effect is unmeasurable.

2.1 A feedback taxonomy by grounding

We classify the reviewer's feedback channel by how much external, task-conditional information it injects:

- **Type 0 – none.** Single-shot, no feedback.
- **Type 1 – ungrounded critique.** An LLM/VLM critiques the artifact from its weights ("looks wrong"). No execution, no measurement.
- **Type 2 – static analysis.** Linters/type-checkers: signal about the artifact's *form*, not its behavior.
- **Type 3 – grounded.** The artifact is exercised and the result observed: **3a** execution (run tests, read tracebacks), **3b** rendered geometry (measure the DOM), **3c** temporal (sample animation frames), **3d** factual (search and verify claims).

2.2 Why grounding breaks the ceiling: a DPI bound

Let $A^\star$ be the (task-determined) correct artifact and A a candidate. The proposer is a channel from its inputs Θ (weights) and prompt X to A. A *reviewer* that only reads A and reasons forms the Markov chain $A^\star \to (\Theta, X) \to A \to R$, where R is its critique. By the data-processing inequality,

$$I(A^\star; R) \;\leq\; I(A^\star; (\Theta, X)),$$

so *no* amount of reasoning or ungrounded review (Type 0–2) can recover more information about $A^\star$ than is already in (Θ, X). This bounds reasoning-only scaling. Sampling is bounded differently but no less tightly: for any finite N, best-of-N can only *select* among the N candidates the proposer actually draws, so even an oracle (grounded) verifier cannot return a correct artifact the proposer is too unlikely to sample within the budget. Its practical ceiling is the high-probability support of the proposer's own output distribution, not a new information channel. Both ceilings appear empirically once we measure them (Figure 2 in Section 4): on hard code, both reasoning-only and oracle best-of-N saturate well short of completion. The inequality bounds the *information channel*, not achievable quality directly; we use it to *motivate* the saturation we then measure and treat as the evidence.

Grounded feedback (Type 3) escapes both ceilings by feeding an environment observation $E = g(A, \text{world})$ (a test outcome, a measured bounding box, a search verdict) back into *generation*. Conditioning the next proposal on E lets the proposer synthesize a candidate it would not otherwise have sampled: formally $I(A^\star; (R, E))$ can exceed $I(A^\star; (\Theta, X))$ because E carries information obtained by *running* the artifact against reality. This is the crucial difference from selection (best-of-N cannot create a missing candidate) and from reasoning (which adds no information), and it is why interaction is a distinct axis rather than a constant-factor speedup.

The symmetric claim: ungrounded evaluation cannot measure the gain. The same inequality applies to the *scorer*. A metric M that is itself a function of (Θ, X) (e.g. a VLM judging a screenshot, whose view of A is a lossy, cropped image) satisfies $I(A^\star; M) \leq I(A^\star; (\Theta, X))$ and cannot certify an improvement that lives in the part of A it cannot observe. Grounding the evaluation (measure the artifact directly) is therefore not optional polish: it is a precondition for *detecting* interaction scaling on any modality whose quality is not fully visible to the default judge. Section 5 shows this is exactly the situation for rendered visual artifacts.

2.3 Predictions

The framework makes three testable predictions, each tested below. **(P1)** Reasoning- and sampling-only scaling saturate strictly below the interaction ceiling (Section 4). **(P2)** Within interaction, more actionable grounded channels lift more, and the *execution* of the artifact, not the extra review pass, is the active ingredient (Section 4). **(P3)** On modalities whose quality is invisible to the default judge, the gain is only detectable with a grounded metric, and an ungrounded reviewer can even regress quality (Section 5).

3 Method: A Budget-Aware Proposer–Reviewer Harness

Architecture. The harness is pure scaffolding around a *frozen* frontier model (Claude Sonnet 4, `claude-sonnet-4-20250514`, temperature 0 unless noted): no fine-tuning, no retrieval, no learned controller. A **proposer** emits an artifact. An **environment** executes/renders/measures it to produce a grounded signal. A **reviewer**, the same model in a diagnostic role, converts that signal into a typed defect list. The proposer then revises. The loop repeats up to an iteration cap and *keeps the best-scoring iteration* (so the reviewed condition can never score below single-shot under the same metric). Single-shot is one proposer call.

The three grounded instruments. Every modality is paired with a deterministic or near-deterministic instrument that both *feeds back* and *scores*:

- **Execution** (`pytest`): exact pass/fail plus the failing assertion and traceback. Each task ships a multi-assertion suite targeting the edge cases that distinguish correct from plausible-but-wrong (e.g. the semantic-version comparator is checked on pre-release precedence, not just major/minor/patch). The deep-spec suite is additionally validated against a reference implementation. A "pass" therefore certifies the specified behavior, not mere lint-cleanliness. And the oracle best-of-N baseline selects against these *same* tests, so its 86.7% ceiling is measured by an identical, strong criterion.
- **DOM geometry**: the artifact is rendered headless and every element's true bounding box is read from the layout engine. We compute, exactly, text-on-text overlap ($\geq$ 6 px), out-of-bounds/clipping, container overflow, document overflow ($>$ 16 px), and, new here, *box-group misalignment* (rows/columns of card or pillar boxes flagged for unequal size, misaligned far edges, or uneven gutters). For scrollable web we score horizontal overflow at desktop (1920) and mobile (375) widths. For animations we probe the DOM at each sampled frame.
- **Native video** (Gemini 3.1 Pro): the rendered clip is judged whole against per-requirement binary checks, a near-grounded substitute where pixels, not a still, are observed.

Table 1. Modality instantiations. Each modality pairs a grounded feedback channel with the instrument that both drives revision and scores the result. "†" marks the four modalities scored by the deterministic DOM-geometry instrument and analyzed jointly in Section 5.

Modality	Type	Grounded instrument	Hard suite (N)
Code	3a	`pytest` pass/fail + traceback	15 (+11 deep-spec)
Academic figures†	3b	DOM geometry + alignment	20
Slides†	3b	DOM geometry + alignment	12 real-paper
Web pages†	3b	DOM geometry, multi-width	20
Animations†	3c	DOM geometry, per-frame	20 SVG/CSS
Video editing	3a+3c	script exec + Gemini-native rubric	15
Deep research	3d	search-grounded factual rubric	15

Where no deterministic instrument exists (flowing web content, factual research), we fall back to a binary per-requirement rubric, flagged as such.

Budget. A run has a token budget B split across proposer and reviewer calls. Section 6 sweeps the split, and Table 1 lists the modality instantiations.

All artifacts are produced under a common *design-principle* generation prompt (proximity, alignment, repetition, contrast, deliberate color) so that single-shot quality is already strong. This makes the headroom we measure a property of genuine task difficulty rather than a weak prompt.

Evaluation protocol. Unless noted, the proposer runs at temperature 0 and each modality is evaluated over three independent on-policy seeds, keeping the best-scoring iteration (cap ≤ 3 for geometry, ≤ 5 for code). The unit of comparison is the (task, seed) pair. For each result we report a two-sided paired sign test over decisive pairs and, for the geometric and code effects, a paired-bootstrap 95% confidence interval (10k resamples) on the effect size.

4 Interaction Is a Distinct Axis Beyond the In-Token-Space Ceiling

Reasoning and sampling saturate while interaction does not (P1). On the 15 hard code tasks at a fixed 20K-token budget we compare four strategies: reasoning-only (extended thinking), best-of-N sampling with an oracle verifier, a single-agent loop, and the proposer–reviewer harness. The two in-token-space axes plateau well short of completion while *every* feedback-iterating strategy approaches a perfect pass-rate (Figure 2, full table in Table 3). The best-of-N comparison is the sharpest: its verifier is an *oracle* (the ground-truth tests), so its ceiling is effectively pass@N. The unsolved tasks never yield a passing sample within the budget, yet the harness solves them. A perfect selector cannot return a candidate the proposer never draws. The loop reaches it by conditioning the next draw on execution feedback. This is the separation the framework predicts, and it is information-theoretic, not a tuning artifact.

4.1 The active ingredient is execution, not review (P2)

Code as a clean case study. Code is the cleanest demonstration because the channel is binary and the score is objective: the harness recovers every first-shot failure with zero regressions, lifting single-shot to a strict 100% across all runs (Table 17). The mechanism is consistent

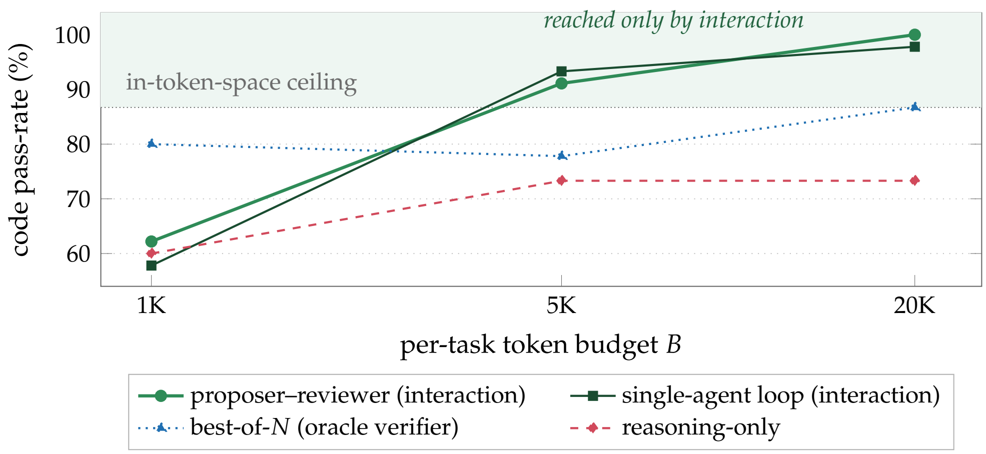


Figure 2. Two in-token-space axes saturate; interaction does not (P1). Measured code pass-rate vs. per-task token budget on the 15 hard tasks (3-seed means, full table with SDs in Table 3). Reasoning-only plateaus at its information ceiling (73.3%) and best-of-N plateaus at 86.7% even with an *oracle* (ground-truth) verifier (dotted): both are functions of the frozen (Θ, X) and flatten with budget, the latter at the high-probability support of the proposer's own output distribution. The two feedback-iterating strategies instead climb past the oracle-sampling ceiling, importing information from execution each cycle—the harness to a strict 100% with *zero* seed variance, the single-agent loop to $\geq$ 97.8% (100% in 2 of 3 seeds). The DPI prediction, made concrete.

across recovered traces. Turn 1 emits code that compiles but mishandles an edge case (off-by-one boundaries, semantic-version comparison, CIDR arithmetic). Turn 2 receives the `pytest` traceback (the failing assertion, the offending input, expected vs. actual) and rewrites the localized block. Turn 3 confirms. Reasoning alone cannot supply this: the model has no way to know its boundary condition is wrong until a concrete test exercises it. A second, harder suite of deep-spec implementations (a from-scratch JSON parser, an A1 spreadsheet-reference resolver, a minimal edit-script diff, an arithmetic/boolean expression evaluator, each validated against a reference solution) corroborates the effect at larger headroom: single-shot 69.7% $\rightarrow$ strict 100% (+30.3 pp, sign-test $p = 0.002$), every failing run recovered with no regressions.

An in-modality control isolates execution. The cross-modal ordering could be confounded by task difficulty, so we fix the task suite and vary *only* the feedback channel: a Type 1 reviewer (LLM cross-review, no execution), a Type 2 reviewer (LLM + `ruff` static analysis, still no execution), and the Type 3a execution baseline. All three arms run *exactly one* reviewing pass, so any separation isolates the *grounding* of the signal, not the act of reviewing (Table 5). The pass-rate gap is modest and we read it cautiously, since it rests on a single task at a single leaner-budget seed. But the *unambiguous* separation is efficiency: execution feedback converges roughly 2.5× cheaper and in fewer iterations, because a concrete failing assertion localizes the fix where an ungrounded critique gropes. The decisive form of this control is the geometry ablation (Section 5), where an ungrounded reviewer not only fails to help but actively *worsens* slide layout: one reviewing pass, opposite sign, set entirely by whether the signal is grounded.[1]

[1]Type 1 and Type 2 do not separate here because the seeded bugs are runtime-logic errors a linter cannot see. Testing the predicted Type 2 > Type 1 ordering would need a `mypy`/`semgrep`-catchable bug set, so the taxonomy's static-analysis tier remains, by our own data, untested.

4.2 Architecture buys efficiency and reliability

Efficiency. Among interaction strategies the gap is not in ceiling pass-rate (they tie within seed noise) but in cost and variance: the proposer–reviewer harness is the most token-efficient and the only variant at a strict 100% with *zero* seed variance, while the single-agent loop reaches 100% in only two of three seeds (Table 6). Context isolation (the reviewer sees only what it needs to localize defects), structured critique (a typed defect list, not raw stderr), and role specialization explain the efficiency.

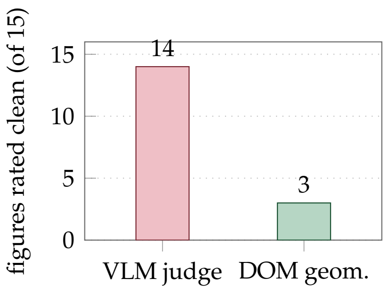
Built-in early stopping. The loop does not over-revise: on code, *every* already-passing single-shot task is submitted at iteration 1 with no revision and no wasted tokens. The grounded signal, a failing test, is the trigger. Without it the loop is silent. This is why code's token overhead over single-shot is modest despite a generous iteration cap.

5 Grounding the Evaluation: The Gain Is Invisible to a VLM

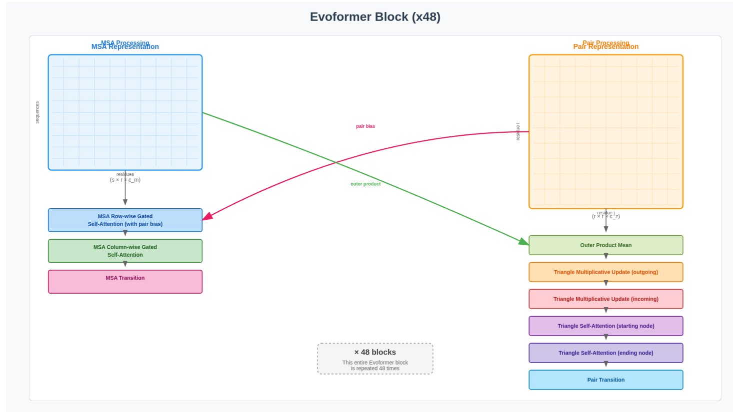
The visual modalities expose the symmetric half of the framework (P3): even when grounded feedback *does* fix the artifact, an ungrounded metric cannot see it.

The default VLM judge is structurally blind. Our starting evaluator is a binary per-requirement rubric scored by a VLM over a screenshot. On 15 dense academic-paper architecture figures, even with native-resolution quadrant tiling, this judge rates 14 **of** 15 single-shot renders "perfect", i.e. satisfying every *content* requirement it is asked to check (the right components present, labels correct). Layout geometry is simply not a question the screenshot lets it answer. By direct inspection the renders are broken (superimposed section titles, labels spilling out of boxes), so the failure is mechanical, not fixable by prompting the judge better. Screenshots are captured at 1080 px, so any content overflowing the canvas is cropped *off-frame before the image reaches the judge* (the defect is literally absent from the pixels the model sees), and small in-frame overlaps fall below its acuity. A deterministic DOM-geometry check finds only 3 **of** 15 are actually clean. The 14-vs-3 gap is therefore not a *wrong* answer to the same question but the judge being *structurally unable to observe* the property that matters (Figure 3a). (This initial probe is a deliberately hard, unconstrained set. The suite scored in Table 2 is generated under the design-principle prompt of Section 3 and is correspondingly cleaner single-shot, with headroom that is genuine task difficulty surviving a strong prompt, not a weak one.)

A deterministic instrument reveals a large, significant effect across four modalities. We score four visual modalities with the same DOM-geometry+alignment instrument under one identical configuration (Sonnet 4, temperature 0, design-principle prompt, three seeds, propose$\rightarrow$measure$\rightarrow$feed-exact-defects-back$\rightarrow$revise, $\leq$ 3 iterations). Table 2 and Figure 5 report the result: grounded geometric feedback produces a large, significant reduction in real layout defects on *all four*, with bootstrap 95% CIs that exclude zero and improvements sharply outnumbering regressions. The magnitudes track single-shot headroom: dense responsive web pages are defect-rich out of the box, while a frontier model under a design-principle prompt already lays out many figures and slides cleanly. Animations have the widest interval because

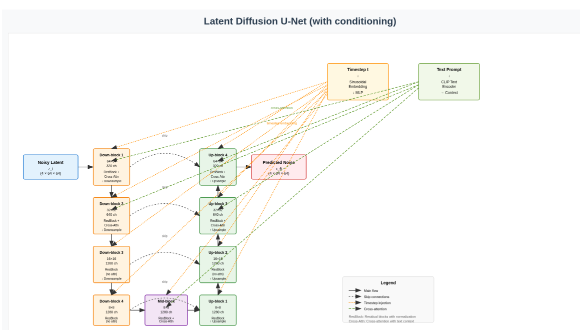


(a) The VLM passes 14/15, the deterministic instrument only 3/15.

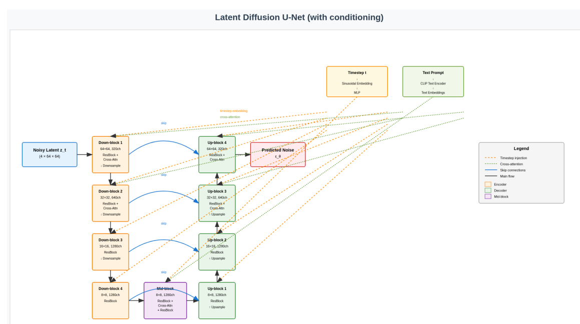


(b) A single-shot figure the VLM rated "perfect": both section titles are superimposed ("MSA Processing" over "MSA Representation"; "Pair Processing" over "Pair Representation") and axis labels collide with the matrix borders.

Figure 3. The default VLM judge is structurally blind to layout defects. (a) On the same 15 single-shot academic figures, the VLM-on-a-screenshot judge rates 14 "perfect" while the deterministic DOM-geometry instrument finds only 3 actually clean. The other 11 are broken in ways cropped off-frame or below VLM acuity. **(b)** A representative "perfect"-rated render whose text-on-text overlaps (≥ 6 px) are exactly what the geometry check flags and the screenshot judge misses.



(a) Single-shot. Conditioning blocks crowd the down-block column, edges route through boxes and tangle at the center, and the legend collides with the decoder, all defects the geometry instrument flags but the VLM cannot see.



(b) After grounded feedback. The same task: conditioning blocks lifted clear, columns spaced and aligned, skip connections deconflicted, legend boxed in free space.

Figure 4. What grounded geometric feedback actually fixes. The same academic-figure task, single-shot (a) vs. after the propose→measure→revise loop (b). This is the per-artifact reality behind the −74% mean defect reduction in Table 2, and exactly the difference a screenshot judge is blind to.

their per-task defect counts are heavy-tailed across the dynamic suite, so we read the sign test as the robust statement and the mean as indicative.

An ungrounded *reviewer* fails to fix layout, and on slides makes it worse (ablation, two modalities). Grounding matters on the feedback side too. We fix the task suite and the proposer model and swap *only* the reviewer's grounding. Because the two arms are separate runs, each is compared against its *own* single-shot baseline, and single-shot geometry carries real seed-to-seed variance. So the load-bearing contrast is the *direction* each reviewer moves its

Table 2. Deterministic geometric-feedback results, one identical configuration across four visual modalities (Sonnet 4, $T=0$, 3 seeds, alignment-inclusive reward). Δ is the mean defect reduction. CI is a paired bootstrap 95% interval on Δ% (10k resamples). p is a two-sided paired sign test. Web is summed over 1920/375 widths, animations over sampled frames.

Modality	n	SS	Rev.	Δ	95% CI	impr./regr. (p)
Academic figures (20)	60	0.57	0.15	−74%	[52, 89]%	17/2 (7×10^{-4})
Dense slides (12)	36	1.25	0.33	−73%	[45, 93]%	13/1 (1.8×10^{-3})
Web pages (20)	60	16.1	8.5	−47%	[30, 62]%	33/2 (4×10^{-8})
Animations (20)	60	16.9	10.2	−40%	[10, 64]%	34/7 (3×10^{-5})

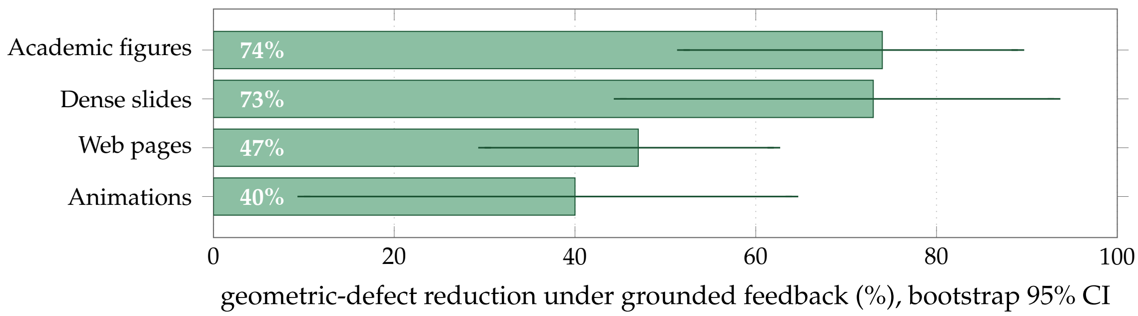


Figure 5. Grounded geometric feedback removes real layout defects on all four visual modalities (mean reduction, with whiskers the paired-bootstrap 95% CIs, all excluding zero). The standard VLM-on-a-screenshot metric measures none of this.

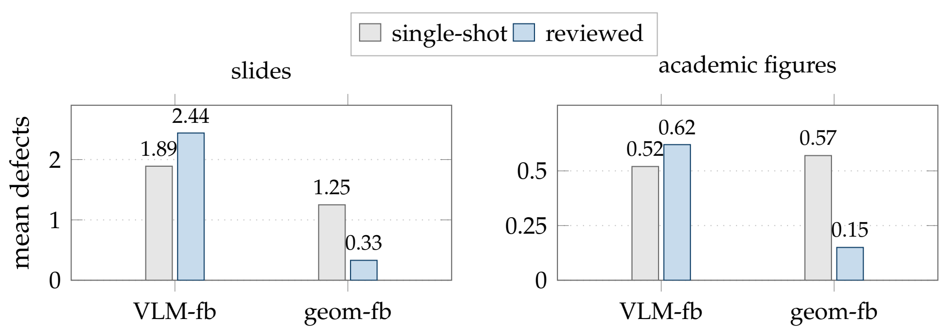


Figure 6. An ungrounded reviewer fails to fix layout, while a grounded one does (controlled ablation, two modalities). Same task suite and proposer model, with only the reviewer's grounding varying. The two arms are separate runs, so each is plotted against its *own* single-shot (hence the differing single-shot bars). The VLM-feedback reviewer (VLM-fb) does not reduce defects and on slides increases them (slides 1.89 → 2.44, figures 0.52 → 0.62, directional/n.s.). The deterministic geometric-feedback reviewer (geom-fb) reduces them (slides 1.25 → 0.33, figures 0.57 → 0.15). The grounded signal is necessary on the feedback side, not just the metric side.

own baseline, not the absolute level (Figure 6). The deterministic geometric-feedback reviewer sharply *reduces* mean defects on both slides and figures, whereas the VLM-feedback reviewer does not: it increases defects on slides and is directionally worse on figures (not significant). Blind to the true geometry, its edits break alignment about as often as they fix it. On the same slides the binary VLM rubric simultaneously saturates, confirming the judge cannot even see the difference. Both halves of Figure 1 are necessary.

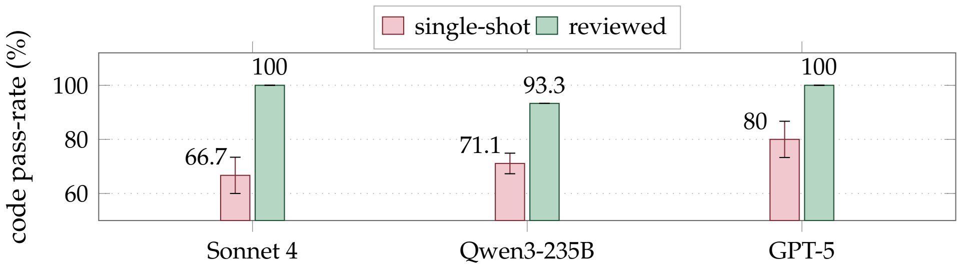


Figure 7. The execution-grounded loop helps every model family. Single-shot vs. harness-reviewed code pass-rate for three model families, each over three independent on-policy seeds (+33.3/+22.2/+20.0 pp). Whiskers are ±1 SD across seeds. The reviewed bars carry *zero* seed variance for all three families.

Is the reduction circular? The instrument both *drives* the revision and *scores* it, and the harness keeps the best-scoring iteration, so *some* reduction is mechanically guaranteed. The honest questions are how large it is and whether it reflects quality a human would care about. Three facts argue it is not merely an artifact of optimizing the reported number. **(i) The defects are real:** single-shot renders carry them at high rate (Figure 3b), and the thresholds are deliberately conservative ($\geq$ 6 px overlap, $>$ 16 px overflow), so a flagged defect is visible, not sub-perceptual. **(ii) The magnitude is not preordained:** keeping the best of three iterations could in principle move the metric by almost nothing. That grounded revision instead removes a large fraction of real defects (Table 2) is a property of the feedback, not the keep-best rule. **(iii) The ungrounded-reviewer ablation is the decisive control:** scored by the *same* metric, the VLM-feedback arm *fails* to reduce it (and worsens slides). So the reduction tracks the *grounding* of the feedback, not the act of optimizing the scorer, a conclusion the model-free cross-model replication (Section 6) reinforces. What remains open, and is the principal caveat of the geometric measurement, is a quantitative human-preference study confirming that DOM-defect reduction maps onto perceived-quality gains across the full suite.

Two modalities where the honest answer is "no headroom." *Video editing*, scored by Gemini 3.1 Pro's native full-clip rubric (a near-grounded instrument that observes pixels, not a still), is already strong single-shot, so the reviewed lift is small and not significant. A hardened multi-step suite de-saturates it. *Deep research*, scored against exact provided facts, saturates: a frontier model knows well-documented facts and does not take the planted traps, so the Type-3d lift is bounded by the near-zero single-shot error rate. This is a genuine limit of the factual modality, since de-saturating it would require facts the judge itself cannot grade. We report these as scoped negatives rather than headline lifts (Table 17).

6 Robustness

Cross-model, code. The effect is architectural, not a Claude artifact. On the 15 hard code tasks, over *three* independent on-policy seeds per family, the harness lifts all three families: Sonnet 4, Qwen3-235B, and GPT-5 (the last from a higher single-shot baseline, Figure 7). The reviewed ceiling has *zero* seed variance for every family while single-shot fluctuates seed-to-seed: the grounded loop converges to the same ceiling regardless of the starting draw.

Cross-model, grounded geometry. To rule out a Claude-specific or prompt-specific explanation of the grounded-evaluation effect, we run the same deterministic geometric-feedback harness with **Gemini 3.1 Pro** as the proposer on the dense real-paper slide suite, scored by the identical model-free DOM instrument. The effect reproduces and is in fact larger: Gemini's single-shot slides are *more* defect-prone than Sonnet's, and the grounded loop removes almost all of the excess (a 93% mean reduction, 19 of 20 decisive task-runs improve). Since the instrument depends on no model, the effect is neither Claude- nor scorer-specific.

Held-out generalization. On a 32-task held-out code suite constructed after the method was fixed, the harness again recovers all first-shot failures with zero regressions and reaches a perfect pass-rate, with no tuning to the original tasks (Table 8).

Budget allocation. Sweeping the proposer/reviewer token split on a fixed budget produces a large pass-rate spread, monotone in the proposer's per-call share (Table 9). The optimum is *propose-heavy*: allocate the majority to generation, reserving enough for the reviewer to localize defects. Extra inference compute is better spent on additional samples than on deeper iteration past the early-stop point.

7 Internalizing the Harness into a Small Student

Two things could carry the harness's gains: the grounded scaffolding, or behavior absorbed into the weights. This section asks how much is the latter: whether the loop's interaction-quality behavior can be *internalized* so a small model emits high-quality artifacts in fewer passes. It is a complement to the thesis, not a substitute: the cheaper internalized proposer is still meant to run *inside* the same grounded harness, so internalization and grounding compose. We distill judge-filtered teacher trajectories from the harness into an 8B Qwen3-VL student via SFT.

Result. The student recovers about half the teacher's mean@1 capability on a 44-task out-of-distribution suite, rising with a second sample (Table 12), and passes a substantial fraction of an 18-task hard held-out suite (Table 11), at roughly 10× lower deployment cost than running the harness over a frontier model (Figure 8). Run back inside the harness, the student recovers still more of the gap.

Variance is a budget (a cautionary finding). Reinforcement fine-tuning (RFT) on top of SFT improves every per-turn consistency metric we measure *and* regresses pass@3 (Table 15). For sampling-based deployment, output-distribution variance *is* the inference-scaling lever, and any post-training that reduces it spends from the same budget best-of-N draws on. Variance is not noise to minimize. It is the resource sampling converts into quality. The operational recipe is therefore: SFT to internalize interaction quality, keep the proposer's sampling temperature, and run it inside the grounded harness.

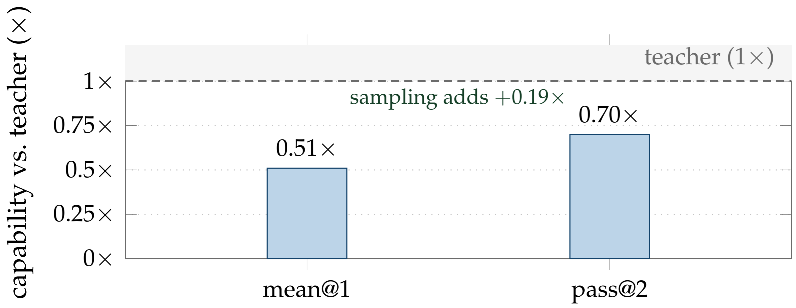


Figure 8. An 8B student internalizes much of the teacher's interaction quality. Distilled 8B markdown student as a fraction of the frontier teacher on the 44-task OOD suite (Table 12): a single sample recovers 0.51× and two samples 0.70× of teacher capability (sampling closes the gap), at ∼10× lower deployment cost. On the 18-task hard held-out suite the same student passes 44%/56% at pass@1/pass@3.

8 Related Work

We now place the framework and results above against the literature on test-time scaling, self-correction, grounded feedback, and model-as-judge evaluation.

Test-time compute scaling. Two axes are established. *Reasoning* scaling (chain-of-thought, tree search, and extended thinking [4, 19, 21, 25, 26]) spends more tokens before committing. *Sampling* scaling (self-consistency and best-of-N [1, 24]) draws multiple attempts and selects. Both transform the same frozen weights and prompt. We make precise (Section 2) the sense in which they are bounded, and position *interaction* as a third axis that imports external information.

Self-correction and critique. Iterative refinement with model-generated feedback (Reflexion [18], ReAct [27], Self-Refine [11], CRITIC [6]) is widely used, but its reliability is contested: Huang et al. [8] show models often cannot self-correct reasoning without an external signal, and Kamoi et al. [9] chart when correction does and does not help. Our taxonomy makes the distinction explicit. *Ungrounded* critique (Type 1) is bounded, whereas refinement against an executed/measured result (Type 3) is not. And our in-modality control (Section 4) shows it is the external execution, not the extra critique pass, that carries the gain.

Grounded and execution feedback. Tool use and execution feedback (Toolformer [16], Gorilla [12], Voyager [23], RLEF [5], and, for website generation specifically, WebGen-Agent's multi-level visual/functional feedback [10]), together with the thinking-vs-doing analysis of Shen et al. [17], establish that acting on an environment helps. We contribute the information-theoretic account of *why*, a feedback-actionability taxonomy that predicts *when*, and, crucially, the requirement that the *evaluation* be grounded too. We also compare harness architectures (single-agent loops [22], sample-and-select [14], proposer–reviewer) under a matched budget.

Model-as-judge reliability. LLM- and VLM-as-judge evaluation is now standard [28], and its failure modes (position, verbosity, and self-preference biases, and broader reward-model fragility [2]) are documented. Our finding is sharper and, to our knowledge, not previously isolated: for artifacts whose quality is a measurable *physical* property (the geometry of a rendered layout), a screenshot judge is not merely biased but *structurally blind* to defects cropped off-frame or below its acuity, so it both fails to score the defect and, used as a reviewer, fails to fix it. This is why a deterministic instrument is necessary, not merely preferable, to detect interaction scaling on visual modalities.

Practitioner evidence at scale. Industrial deployment independently corroborates the gap our framework targets. Hong [7] reports a controlled study at ByteDance (3 frontier coding models $\times$ 3 agent frameworks $\times$ 100 runs, 900 total) on a single real product feature under a fixed prompt. Functional *correctness* exceeded 80% for every model–framework pair, yet "deliverability" axes (UI usability, reliability, maintainability, performance, compatibility) collapsed to a failing 40–60% with high run-to-run randomness. Only after adding harness "infrastructure" (context engineering, architectural constraints, and persisted team memory) did deliverability rise to $\sim$80%, while correctness barely moved (80%$\rightarrow$90%). The same report notes that despite $>$90% of one team's code being model-written, throughput rose only 1.6$\times$, far short of the raw generation speedup. That shortfall is the macroscopic shadow of the quality single-shot generation does not deliver. This is the signature we formalize and measure: model and sampling scaling saturate the easy-to-see axis, while the interaction loop carries the rest. Tellingly, the report had to define a multi-axis quality metric before the gap was visible at all, an industrial analogue of our grounded-evaluation requirement. Its grounded fixes (e.g. browser-driven validation before commit) are Type-3 feedback in our taxonomy, not ungrounded self-review.

Information theory. The data-processing inequality [3] is the lens for our bound: any post-processing of the model's outputs (reasoning, ungrounded review) cannot increase information about the target, whereas an environment observation introduces a new channel.

9 Discussion and Conclusion

Grounding is the load-bearing variable. Two of the three test-time axes are bounded by the data-processing inequality because they never leave the model's token space. The third is not bounded by that ceiling, because grounded feedback imports external information each cycle (it is of course still bounded, by the information the environment exposes and the proposer's ability to act on it). We have argued and shown that this grounding must hold on *both* sides of the loop. On the feedback side, an in-modality control isolates test *execution*, not the extra review pass, as the active ingredient, and a VLM *reviewer* of slide layout actively regresses quality. On the evaluation side, the default VLM-on-a-screenshot metric is structurally blind to the geometric defects the loop fixes, so the entire visual-modality effect is invisible until the metric is grounded in deterministic DOM geometry. Once it is, grounded feedback removes a large, significant fraction of real layout defects across four modalities, with every CI excluding zero (Table 2).

A methodological warning for the field. Much of the recent excitement about "LLMs that critique and improve their own visual output" is measured with VLM judges. Our results imply many such measurements are uninformative on layout: the judge cannot see the defects, and using it as the reviewer can make things worse. Where an artifact's quality is a measurable physical property (geometry, execution, retrieval), the evaluation should be grounded in that property, not in a model's picture of it.

Grounding is the theory under "loop engineering." Practitioners have converged on an informal art of *loop engineering*, or *loopcraft* (stacking an inner agent loop with an outer *verification* loop that critiques and revises), as the design pattern for capable agents [15, 20]. That verification loop is exactly the proposer–reviewer loop we study, and our results supply the variable the practitioner framing leaves implicit: *grounding* decides when stacking such a loop pays off. A verification loop closed on an ungrounded signal (model critique, or a VLM reading a screenshot) is bounded by the data-processing inequality (Section 2) and, on visual artifacts, actively harmful: its edits make layout geometry *worse* (Section 5). Only a loop closed on an executed/rendered/measured observation imports information and clears the reasoning/sampling ceiling. The lesson is not "add more loops" but "ground the loop you add," on both the feedback and the evaluation side.

Limits, honestly stated. Two modalities saturate: video editing is already strong single-shot (the lift is real only on a hardened multi-step suite), and deep research saturates because frontier models know well-documented facts and the judge cannot grade facts they do not. Deterministic geometry applies to artifacts with a measurable intended layout (figures, slides, web, animations). For genuinely flowing content we still rely on a binary rubric, with the VLM-reliability caveat. And because the geometric instrument both supplies the feedback and scores the result, the reported defect reductions are partly an optimization of the metric itself. We argue in Section 5 (via conservative thresholds, the non-trivial magnitude, and the ungrounded-reviewer control that moves the same metric the wrong way) that they nonetheless reflect real, human-visible defects, but a quantitative human-preference validation remains future work.

Future work and broader impact. Two directions follow directly. First, deterministic instruments for modalities beyond layout and execution (differentiable or symbolic checks for audio, 3D, tabular, and UI-interaction artifacts) would extend grounded evaluation to where VLM/LLM judges are likewise blind. Second, our DPI argument bounds an information channel. Tightening it into a quantitative relationship between feedback fidelity and achievable quality is open. On impact: grounding the metric is a cheap, deployable correction to a measurement error that, we argue, currently inflates reported progress on self-improving visual generation. The same instruments are auditable and reduce reliance on opaque model judges. The harness itself is scaffolding over a frozen model, so it inherits that model's capabilities and failure modes rather than introducing new ones.

Operational recipe. Wrap a frozen frontier model in a proposer–reviewer harness with the most actionable grounded feedback channel for the modality. For visual artifacts, ground *both* the reviewer and the scorer in deterministic geometry: measure the rendered DOM, never a screenshot. Allocate the majority of the per-task budget to the proposer, and spend extra compute on samples rather than deeper iteration. For cost-sensitive deployment, distill the proposer into a small student and run it inside the same harness. Interaction scaling is real, distinct from reasoning and sampling, and predictable from the feedback taxonomy. Once you ground the metric, it is plainly visible.

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A Detailed Results and Configurations

This appendix collects the full tables underlying the summaries in the main text, from the four-strategy scaling curves through the distillation ablations to the original six-modality holistic overview (the last superseded for the visual modalities by the grounded measurements of Section 5).

A.1 Four-strategy scaling at a matched 20K-token budget

Table 3. The headline scaling result (3-seed mean ± SD). Pass-rate by strategy × budget on 15 hard code tasks, Sonnet 4. S, L, and H are averaged over three independent on-policy runs. R was run once (extended-thinking variation across seeds is small at temp 0) at the sweep's budget partition (a more generous partition lifts R to 86.7%, see footnote and Table 4). S's small dip from B=1K (80.0%) to B=5K (77.8%) is within the ±3.1 pp seed noise on a 3-seed × 15-task suite, since pass@N is non-decreasing in N in expectation but not pointwise across finite-sample on-policy seeds. *Reasoning-only and best-of-N saturate well below 100%*, while all three iteration-with-feedback strategies reach ≥ 97.8% at B=20K (L hit 100% on 2 of 3 seeds, H on all 3, so H is the only variant at strict 100% across seeds). The architectural ordering across H, L is within seed noise on pass-rate. What separates them is token efficiency and reliability (Section 4).

Budget B	R (reasoning-only)	S (best-of-N)	L (single-agent loop)	H (proposer–reviewer)
1K	60.0%	80.0% ±0.0 pp	57.8% ±3.1 pp	62.2% ±3.1 pp
5K	73.3%	77.8% ±3.1 pp	**93.3%** ±0.0 **pp**	91.1% ±3.1 pp
20K	73.3%	86.7% ±0.0 pp	97.8% ±3.1 pp	**100.0%** ±0.0 **pp**
Ceiling	73.3%	86.7%	∼100%	**100% (zero variance)**

A.2 Reasoning-only at a matched budget

Table 4. Reasoning-only at matched budget vs. single-shot and the harness on the 15 hard code tasks. Single-seed numbers (the Phase-1 reference seed, also used in Table 5). The 3-run aggregate is reported in Table 17 ($66.7 \pm 6.7\%$ single-shot, $100.0 \pm 0.0\%$ reviewed). *Reading the numbers:* single-shot here is this single seed's $11/15 = 73.3\%$ (the canonical 3-seed mean is 66.7%), and the reasoning value shown is the *thinking-heavy* partition (86.7%). This coincides numerically with the oracle best-of-N ceiling, while the *default*-partition reasoning ceiling (73.3%, Table 3) coincides with this seed's single-shot. Both are coincidences of the small suite, not the same run. Reasoning closes two-thirds of the harness's gap over single-shot but falls 6.7 pp short of the harness's pass-rate even at 1.57× the harness's token budget.

Strategy	Pass-rate	Mean tokens	Δ vs. single-shot
Single-shot	73.3% (11/15)	1,406	—
Reasoning-only (thinking-heavy partition)	86.7% (13/15)	4,137	+13.3 pp
Proposer–reviewer harness	**93.3% (14/15)**	**2,643**	+**20.0** pp

A.3 In-modality feedback-type control (Type 1/2/3a) on a fixed code suite

Table 5. In-modality feedback-type controls on 15 hard code tasks (Sonnet 4 proposer and reviewer). Each row is a single on-policy seed. The "SS" column is the *per-seed* single-shot pass-rate (not held constant across rows) and "Reviewed" is the ceiling after that seed's harness run. The three SS values (10/15 and 11/15) are draws from the same single-shot distribution that gives Table 17's 3-run mean of $66.7 \pm 6.7\%$ (i.e. within the ± 1 SD band centered on 10/15). Because the SS column varies by row, the load-bearing comparison is the *reviewed* ceiling: Type 3a reaches 14/15, Type 1/2 reach 13/15 (+6.7 pp), *and* Type 3a is ∼2.5× more token-efficient (2,643 vs. ∼7,100 tokens/task, 1.53 vs. 2.07 iterations). The single task unrecovered in every control row is `code_011` (CJK text wrapping). The full harness at B=20K recovers it across all three seeds (Table 3), which is why the headline reviewed rate is the strict 100% of Table 17 while this leaner single-seed control caps at 14/15.

Type	Reviewer sees	SS	Reviewed	Iters
None (single-shot)	—	11/15	—	—
Type 1 (cross-review)	code only	10/15	13/15	2.07
Type 2 (+`ruff`)	code + static analysis	11/15	13/15	2.07
Type 3a (+exec)	code + test stderr/traceback	11/15	**14/15**	**1.53**

A.4 Single-agent loop vs. proposer–reviewer harness

Table 6. Three interaction-strategy variants at B=20K on the 15 hard code tasks, all using Type 3a execution feedback. Pass-rate is within seed noise (sign-test p>0.6 on the H–L comparison). The harness wins on *token efficiency* (1,029 vs. 1,431 L vs. 1,416 IAD mean output tokens) and on *seed variance* (zero across three seeds for H, while L hit 100% in 2 of 3 seeds and IAD was run for one seed only).

Strategy	Pass-rate	Mean tokens	Notes
L (single-agent loop)	97.8% ±3.1 pp (3 seeds)	1,431	1 task in 1 seed
IAD (oracle, K=3, T=0.7)	100.0% (1 seed)	1,416	+38% tokens vs. H
H (proposer–reviewer)	**100.0%** ±0.0 **pp (3 seeds)**	**1,029**	zero seed variance

A.5 Cross-model replication (code)

Table 7. Cross-model replication on the same 15 hard code tasks, each family over *three* independent on-policy seeds at T=0.7 (Claude is the multi-run reference from Table 17). The reviewed ceiling has zero seed variance for all three families, and the harness lift survives across Anthropic, Alibaba, and OpenAI. Single-shot per-seed pass-rates: Qwen 66.7/73.3/73.3%, GPT-5 86.7/80.0/73.3%.

Model	Single-shot pass	Reviewed pass	Δ (lift)
Claude Sonnet 4 (3 seeds)	66.7 ± 6.7%	**100.0 ± 0.0%**	+**33.3** pp
Qwen3-235B-Instruct-2507 (3 seeds)	71.1 ± 3.8%	**93.3 ± 0.0%**	+**22.2** pp
GPT-5 (3 seeds)	80.0 ± 6.7%	**100.0 ± 0.0%**	+**20.0** pp

A.6 Held-out generalization (code)

Table 8. Held-out generalization on 32 newly constructed code tasks (zero overlap with the 15-task development set). The harness recovers 3/3 single-shot failures and regresses zero passing tasks. The smaller absolute Δ is a baseline-compression effect (single-shot is 90.6% on the held-out set, leaving at most 9.4 pp headroom).

Split	N	Single-shot	Reviewed	Δ	SS-fail fixes / regr.
Development (3-run mean)	15	66.7 ± 6.7%	100.0 ± 0.0%	+33.3 pp	5/5 / 0
Held-out v2	32	90.6%	**100.0**%	+**9.4** pp	**3/3 / 0**

A.7 Budget-allocation simplex

Table 9. Budget allocation simplex on 15 hard code tasks, $B = 10\text{K}$ total. Pass-rate is monotone in the proposer's per-call cap, and review-heavy corners (low b_1) collapse. The 9-point sweep yields an 86.6 pp spread.

Allocation	b_1 (prop)	b_2 (exec)	b_3 (rev)	Pass-rate	Mean tokens
A (propose-heavy)	0.80	0.10	0.10	**93.3% (14/15)**	3,101
G (prop-dominant)	0.50	0.25	0.25	**93.3% (14/15)**	2,697
D (prop+exec)	0.40	0.40	0.20	80.0% (12/15)	4,025
E (prop+review)	0.40	0.20	0.40	80.0% (12/15)	4,388
I (equal)	0.33	0.34	0.33	66.7% (10/15)	4,845
H (review-dominant)	0.25	0.25	0.50	40.0% (6/15)	7,485
F (exec+review)	0.20	0.40	0.40	13.3% (2/15)	8,771
B (execute-heavy)	0.10	0.80	0.10	6.7% (1/15)	9,383
C (review-heavy)	0.10	0.10	0.80	6.7% (1/15)	9,383
			Spread:	**86.6 pp**	

A.8 Token ROI by modality

Table 10. Per-task token cost and quality return-on-investment for the harness, averaged across three on-policy runs. "Extra tokens" is reviewed-minus-single-shot per task. Tokens per 0.01 quality is extra tokens divided by (mean reviewed quality − mean single-shot quality)·100. Code dominates by 100×.

Modality	Single-shot tokens	Reviewed tokens	Extra tokens	Tokens / 0.01 quality
Code	3,336	4,575	1,239	**37**
Video	3,808	20,584	16,776	359
Research	14,116	27,793	13,677	2,415
Webpages	14,060	80,148	66,088	5,128
Animations	25,491	89,009	63,518	4,331
Slides	10,274	36,956	26,682	4,447

A.9 Distillation: headline

Table 11. Distilled 8B student on the 18-task hard held-out suite (teacher's pre-curated single-shot failures, so teacher single-shot is 0/18 by construction). The student lifts judge-keep from 0% to 44% at pass@1 and 56% at pass@3. The 0.51× mean@1 capability ratio against the teacher is established on a separate 44-task out-of-distribution suite (Table 12), where teacher single-shot is non-degenerate.

Setup	Compute	Judge-keep
1 sample × 5 turns	1×	44% (8/18)
1 sample × 8 turns	1.6×	28% (regressed)
3 samples × 5 turns (pass@3)	3×	**56% (10/18)**

A.10 Distillation: out-of-distribution capability ratio

Table 12. Distilled 8B student (V3 + force-close) on a 44-task out-of-distribution suite where the teacher's single-shot is non-degenerate (teacher 20/44). Capability ratio is the student's judge-keep divided by the teacher's 20/44. Two sampling seeds are used, and pass@2 is the union over both. This is the suite that establishes the ∼0.51× mean@1 / 0.70× pass@2 ratios quoted in Section 7 and Figure 8.

Metric	sample 1	sample 2	mean@1	pass@2
Judge-keep	9/44 (20%)	11/44 (25%)	23%	**32**%
Capability ratio (vs teacher 20/44)	0.45×	0.55×	0.51×	**0.70**×

A.11 Distillation: capability U-curve over force-close budget

Table 13. Reasoning-cap U-curve on the 18-task hard held-out suite. SFT recipe sweep on Qwen3-VL-8B-Instruct (37 judge-kept teacher traces, same inference protocol). The optimum is at the student's coherence horizon, not the teacher's.

Variant	Reasoning cap (chars)	Judge-keep	Identical-retry
V1 (no reasoning)	0	6%	67%
V4	1000	11%	35%
V2	3000	17%	32%
V3 (chosen)	**1500**	**44%**	10%

A.12 Distillation: data-scale

Table 14. Three independent attempts to scale the SFT corpus past V3's 37 examples. All regressed. Quality dominates quantity at this scale.

Variant	Composition	n examples	Judge-keep
V3 (chosen)	235B teacher, judge-kept	37	**44%**
V5	V3 + 13 judge-rejected (235B)	50	33%
V6	V3 + 20 judge-kept (30B teacher)	57	33%
RFT v1	Self-rolled, judge-filtered	61	39%

A.13 Distillation: RFT regresses pass@3

Table 15. RFT polishes consistency at the cost of pass@k. Per-turn metrics improve, per-task variance collapses, and pass@3 regresses by 17 pp. For deployment with sampling, the model's variance *is* the inference-scaling budget. *Note on notation.* **mean@1** is the expected pass-rate of a single random sample, computed as the mean accuracy across the three sampling seeds (one T=0 greedy + two T=0.7). This is the natural variance-sensitive metric and differs from the greedy single-seed *pass@1*=44% reported in Table 11, which is just seed 1's accuracy. *pass@2* and *pass@3* are union pass-rates over the first 2 / all 3 seeds.

	Consistency metrics			Sampling-aware judge-keep		
	id-retry ↓	addresses-rev ↑	no-artifact ↓	mean@1	pass@2	pass@3
V3 (chosen)	9%	83%	2	**31%**	**50%**	**56%**
RFT v1	**8%**	**89%**	**1**	28%	39%	39%

A.14 Per-task pass@k for the 8B markdown student

Table 16. Per-task judge-keep outcome for the 8B markdown student on the 18-task hard held-out suite. ✓ = judge-kept, · = judge-rejected.

Category	Task	seed 1 (T=0)	seed 2 (T=0.7)	seed 3 (T=0.7)	pass@3
webpages	web_002	✓	·	·	✓
	web_003	✓	·	✓	✓
	web_005	·	·	·	·
	web_008	·	·	·	·
	web_012	·	·	·	·
slides	slide_001	·	·	·	·
	slide_006	·	·	·	·
	slide_007	·	·	·	·
	slide_008	✓	·	·	✓
	slide_010	·	·	·	·
	slide_014	✓	✓	✓	✓
	slide_018	✓	✓	·	✓
	slide_020	·	✓	·	✓
code	code_h001	✓	·	·	✓
	code_h002	✓	✓	·	✓
	code_h003	·	·	·	·
	code_h004	·	·	✓	✓
	code_h005	✓	✓	✓	✓
	Total kept:	8/18	5/18	4/18	**10/18 (56%)**

A.15 Original six-modality holistic overview (historical)

Table 17. Single-shot vs. reviewed quality across six modalities, Claude Sonnet 4. Means and standard deviations are computed across three independent on-policy runs (per-run means, then aggregated). Code is binary pass-rate, and other modalities are continuous in $[0, 1]$. p-values: one-sided paired sign test on per-task means. **Bold** marks rows where the lift clears $p < 0.05$. Code is scored by execution (`pytest`), video by a native full-video rubric (Gemini 3.1 Pro judging the whole rendered clip), and the four visual/factual rows use a holistic screenshot judge and serve only as a cross-modal overview. *This table is historical and is superseded for the visual modalities by the grounded measurements of Section 5*: under the deterministic DOM-geometry instrument (Table 2) the figure, slide, web, and animation effects are large and significant, whereas the holistic-judge numbers here cannot see the geometric defects and therefore mis-state them. The visual rows here also use the earlier, smaller task sets (15 web and 15 animation tasks, 20 generic slides), not the hardened suites of Table 2 (20/20/12 real-paper), which is why the per-modality N differs between the two tables. The deep-research and video rows also use *different* instruments than Section 5. Research here is the old holistic judge (0.63 $\to$ 0.69) rather than the search-grounded factual rubric (single-shot 0.982), so the large numeric gap is an instrument change, not a contradiction. Video uses the same Gemini-native rubric in both places (0.70 $\to$ 0.74).

Modality	N	Feedback type	Single-shot	Reviewed	Δ (sign-test p)
Code	15	3a	$66.7 \pm 6.7\%$	$\mathbf{100.0} \pm \mathbf{0.0}\%$	$+\mathbf{33.3}$ pp (0.008)
Video editing	15	3a+3c	0.70 ± 0.03	0.74 ± 0.05	+0.04 (0.64)
Deep research	15	3d	0.63 ± 0.06	0.69 ± 0.06	+0.06 (0.090)
Web pages	15	3b	0.43 ± 0.03	0.56 ± 0.01	+0.13 (0.073)
Animations	15	3c	0.40 ± 0.03	0.55 ± 0.01	+0.15 (0.073)
Slide decks	20	3b	0.73 ± 0.01	0.79 ± 0.01	+0.06 (0.090)